Choices You Make Should Matter: Asynchronous Discovery Activities

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Background

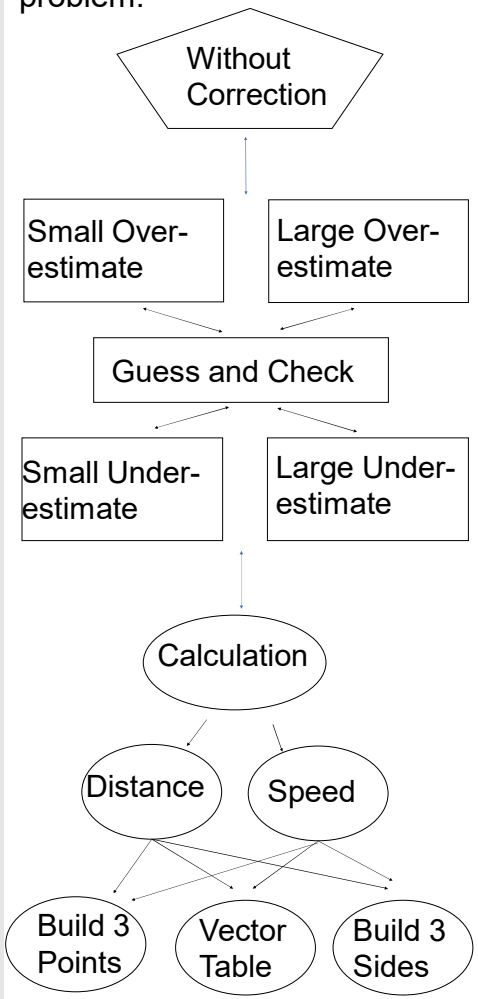
- COVID-19 produced a massive shift in the use of online modalities to teach math
- There are limited examples of digital discovery-based activities
- Asynchronous activities need to consider both students' mental constructions as well as non-content motivations and goals

Methods

- Data collected over a period of 2 years
- Activity went through 3 design changes:
  - **Static:** Traditional Law of Sines/Cosines and/or Vectors word problem
  - **Static with Viz:** Traditional word problem with applet that visualizes flight paths
  - **Dynamic Responses:** Branching choices and stepwise validation of questions/options presented to students
- Analyzed student answers and how they interacted with Viz + Dynamic activity

Takeaway

Conflict points acted as **nodes** in a **directed graph** that provided **non-linear paths** for students to **explore** a word problem.



Results

- **Primary Conflict** (Solution vs Understanding) – Primary use was as a tool to solve the problem, not as a tool to explore the problem
- **Secondary Conflict** (Real-World Validation) – Some students used career expertise to estimate solution without engaging with the "math"
- **Tertiary Conflict** (Extreme Conditions) – Need for a dynamic response that follows the student's logic and contradictions their expectations

Conclusions

- Applet reflected common methods students used to solve word problem
- More work to analyze why students made choices

Try the activity here!